

# Programming for Machine Learning and Data Science

## Semester-end Examination – Part 1

May 23, 2026: 10:00 – 12:00

Total Marks: 30

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### INSTRUCTIONS

This examination has two parts:

1. Part-1, 30 marks: 10:00 am – 12:00 pm (**this question paper**)
2. Part-2, 20 marks: 12:15 pm – 1:15 pm (**question paper will be distributed at 12:15 pm**)

#### Instructions for Part-1

- This part is a **paper-less** examination. Your solution files (see below) should be **ZIPped** and uploaded to Olympus, **by 12:10 pm**
- Solutions should be created using coding + documentation environment like Google Colab / Jupyter Notebook / etc.
- You are recommended to use the Notebook itself for documenting your explanations / observations / analysis / conclusions.
  - *However, if you so desire, you can submit a separate document (PDF) containing your analysis and report.*
- **Note:** In case you use online tools such as Google Colab, be sure to download and submit '.ipynb' file(s) and **NOT** links to the online Notebook. Likewise, if you create any (report) document online, submit its PDF.
- File names of the Notebook / PDF files SHOULD have the following format "**FULL NAME\_EOID**".
- The Notebooks should flawlessly execute with your data file(s) kept in the same folder.
- Upload the Notebooks, PDF of your report (if separately created) to Olympus, to "**Endsem-Part-1**"
- You can access your notes, slides, Internet based tools (**including code generation tools**) during the test. However, the following conditions and restrictions will strictly apply:
  - **You will be deemed to be the creator of the code. This requires you to completely own and understand the generated code and, if required, explain it later.**
  - **AUTOMATIC GENERATION OF OVERVIEWS / SUMMARIES / REPORTS / ANALYSIS / CONCLUSIONS WILL ATTRACT PENALTIES and MAY LEAD TO REJECTION OF THE SUBMISSION AND AWARD OF ZERO MARKS. All these should be your own!**
  - Copying, collaboration, use of ANY online collaboration tool / other means of collaboration is **NOT ALLOWED** at any time.

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**Under no circumstances will submissions be accepted after 12:10 Hrs.**

In case of any queries related to these instructions, contact an invigilator

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Attempt all questions. It is very important to apply Data Science best practices while solving these problems.

| QNo | Questions and Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <p>The dataset <b>endsem-2026-05-set-1-final.csv</b> is sourced from satellite images of agricultural areas. In addition to the location coordinates, the data originally contained an integer column through which each location was associated with a specific crop code. A collection of observations forms a crop area. Sadly, this information has been lost during data handling. However, it is believed that the crop areas can still be reconstructed from the given data. It is your task to salvage this information and <b>create prediction model(s)</b>, such that given the location coordinates, the model predicts the crop.</p> <p>(In your solution clearly identify the question numbers being attempted: 1(a), 1(b), etc.)</p> <ul style="list-style-type: none"><li>a) Document your understanding of the data and the problem(s) you need to solve to accomplish the given task [3 marks]</li><li>b) Explain, with reasons, the steps you will take to make the data ready for creating prediction model(s) [3 marks]</li><li>c) Implement these steps, and analyze the outcomes of these steps [3 marks]</li><li>d) Explain, with adequate reasons, which prediction model(s) you will prefer to create [3 marks]</li><li>e) Create the model(s), analyze them, and state and justify your conclusions [3 marks]</li></ul> <p style="text-align: right;"><b>Total Marks: 15</b></p> |
| 2   | <p>Use the dataset <b>endsem-2026-05-set-2-final.csv</b> to answer the following questions:</p> <ul style="list-style-type: none"><li>a) Describe the data, and its peculiarities [3 marks]</li><li>b) You are required to create <b>parametric regression model(s)</b> using this data set to achieve an <math>R^2</math> value of at least 0.75. Explain your strategy to achieve this goal [3 marks]</li><li>c) Create the model(s) and analyze them [3 marks]</li><li>d) Create at least one <b>non-parametric regression model</b> to achieve significantly better results [3 marks]</li><li>e) Explain, with adequate reasons, the differences you observe between the models you have created [3 marks]</li></ul> <p style="text-align: right;"><b>Total Marks: 15</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |